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|------------------------|---------------------------------------------------------------------------------------------------|
| <b>Practical : 1</b>   | <b>Set up Python environment for AI and implement a simple reflex agent in a toy environment.</b> |
| <b>Date : 05/08/26</b> |                                                                                                   |

**Aim: -**

To set up a Python environment for Artificial Intelligence and implement a simple goal-based agent in a toy environment. The agent navigates through a grid environment, cleans all dirty locations, and finally reaches the goal location.

**Algorithm:-**

1. Start the program.
2. Import the required libraries: random, time, os, and datetime.
3. Display the program title.
4. Display the current date, time, and operating system.
5. Randomly generate the room status as **Dirty** or **Clean**.
6. Check the room status.
7. If the room is **Dirty**, start cleaning the room.
8. Otherwise, display that no cleaning is required.
9. Display the task completion message.
10. Stop the program.

### **Program:-**

```
# =====  
# Practical No. 1  
# Simple Reflex Agent  
# =====  
  
import random  
import time  
import os  
import datetime  
  
print("=" * 60)  
print(" SIMPLE REFLEX AGENT")  
print("=" * 60)  
  
current_time = datetime.datetime.now()  
  
print("\nCurrent Date :", current_time.strftime("%d-%m-%Y"))  
print("Current Time :", current_time.strftime("%H:%M:%S"))  
  
print("\nOperating System :", os.name)  
  
room_status = random.choice(["Dirty", "Clean"])  
  
print("\nChecking Room Status...")  
  
time.sleep(2)  
  
print("\nRoom Status :", room_status)  
if room_status == "Dirty":  
    print("\nAI Decision : Start Cleaning")  
    time.sleep(2)  
    print("Action : Room Cleaned Successfully")  
else:  
    print("\nAI Decision : No Cleaning Required")  
    print("Action : Room is Already Clean")  
  
print("\nTask Completed Successfully.")  
print("=" * 60)
```

### Output:-

```
C:\Users\Vishnu>cd C:\Users\Vishnu\OneDrive\AI_Practicals

C:\Users\Vishnu\OneDrive\AI_Practicals>python practical1_reflex.py
=====
SIMPLE REFLEX AGENT
=====

Current Date : 23-08-2026
Current Time : 21:00:17

Operating System : nt

Checking Room Status...

Room Status : Dirty

AI Decision : Start Cleaning
Action : Room Cleaned Successfully

Task Completed Successfully.
=====
```

```
C:\Users\Vishnu\OneDrive\AI_Practicals>python practical1_reflex.py
=====
SIMPLE REFLEX AGENT
=====

Current Date : 23-08-2026
Current Time : 21:49:15

Operating System : nt

Checking Room Status...

Room Status : Clean

AI Decision : No Cleaning Required
Action : Room is Already Clean

Task Completed Successfully.
=====
```

**Conclusion:-** The Python environment was successfully set up and Thus, the **Simple Reflex Agent was successfully implemented using Python**. The agent observes the current room status (**Dirty or Clean**) and takes an immediate action based on predefined rules. If the room is dirty, the agent cleans it; if the room is clean, no cleaning action is required.